Curriculum Vitae

Xuan Feng

Gender: Male
Birthdate: January 1996
Telephone: +65-96137864
E-mail: xuan.feng0114@gmail.com



Education

Ph.D.: Polymer chemistry and physics, Changchun Institute of Applied Chemistry, Chinese Academy of Sciences (University of Science and Technology of China), Changchun, PR China, 2017-2022

B.Eng.: Polymer Materials Science and Engineering, Ocean University of China, Qingdao, PR China, 2013-2017

Work Experience

Research Fellow: School of Chemistry, Chemical Engineering and Biotechnology, Nanyang Technological University, Singapore, 2022-present

Honors and Awards

Shandong Weigao Group Scholarship, Changchun Institute of Applied Chemistry, 2021

Postgraduate Scholarship, University of Science and Technology of China, 2017-2020

School-level comprehensive scholarship, Ocean University of China, 2015-2016

School-level physical and mental quality scholarship, Ocean University of China, 2014

Research Topics

Equilibrium self-assembly of anisotropic colloids

Glass transition in colloidal systems

Assembly and disassembly of polymers

Computational methods

Carbohydrate polymer simulations

Publications

1. **Feng, X.**, Li, F.*, Ding, M., Zhang, R.* & Shi, T. (2020). Molecular dynamic simulation: Conformational properties of single-stranded curdlan in aqueous solution. *Carbohydrate Polymers*, 250, 116906.
2. **Feng, X.**, Li, F., Ding, M. M., Zhang, R.*, Shi, T. F.* & Jiang, W. (2021). Molecular dynamic simulation: Structural insights of multi-stranded curdlan in aqueous solution. *Carbohydrate Polymers*, 261, 117844.
3. **Feng, X.**, Li, F., Ding, M. M., Zhang, R.*, Shi, T. F.*, Lu, Y. & Jiang, W. (2022). Molecular dynamic simulation: Study on the recognition mechanism of linear β-(1→3)-D-glucan by Dectin-1. *Carbohydrate Polymers*, 286, 119276.
4. **Feng, X.**, Yan, N., Jin, J.* & Jiang, W.* (2023). Disassembly of Amphiphilic AB Block Copolymer Vesicles in Selective Solvents: A Molecular Dynamics Simulation Study. *Macromolecules*, 56 (6), 2560-2567.
5. Zhang, H.#, **Feng, X.**#, Xia, X., Zhu, J., Wang, H.*, Ni, R.* & Zhang, Z.* (2023). Shape-Dictated Self-Assembly of Photoresponsive Hybrid Colloids. *Small Methods*, 7 (9), 2300383.
6. Zhang, T.#, Lyu, D.#, Xu, W.#, **Feng, X.**, Ni, R.* & Wang, Y.* (2023). Janus particles with tunable patch symmetry and their assembly into chiral colloidal clusters. *Nature Communications*, 14, 8494.
7. Gao, Y., **Feng, X.***, Zhang, R., Xiao, J., Huang, Q., Li, J. & Shi, T.* (2024). Molecular dynamics simulation: Effect of sulfation on the structure of curdlan triple helix in aqueous solution. *International Journal of Biological Macromolecules*, 282, 137119.
8. Liang, J.#, **Feng, X.**#, Zheng, N., Wang, H.*, Ni, R.* & Zhang, Z.* (2025). Glass Transition in Monolayers of Rough Colloidal Ellipsoids. *Physical Review Letters*, 134 (3), 038202.
9. Zhong, B., **Feng, X.**, Gao, Y., Zhan, S. & Shi, T.* (2026). Recognition Mechanism of Dectin-1 for Phosphorylated β-(1→3)-D-Glucan in Potential Recognition Regions Based on Molecular Dynamics Simulations. *Chemical Journal of Chinese Universities*, 47 (6), 206-217.